

IJP Notes Collection

Topic Index

A guided map of the complete 21-day collection, from IBM role preparation and core AI foundations through enterprise GenAI solution design.

21	480	27,682
TOPIC NOTES	PDF PAGES	SOURCE LINES

RECOMMENDED PATH

- 1

Core AI and data foundations
Days 1-7: IBM role expectations, ML, Python, EDA, NLP, LLMs, and databases.
- 2

RAG and agent workflows
Days 8-14: RAG, embeddings, evaluation, LangChain, LangGraph, and multi-agent AI.
- 3

Data platform and production delivery
Days 15-21: Databricks, Delta Lake, MLflow, MLOps, APIs, cloud, and enterprise GenAI.

Core AI and data foundations

IBM role expectations, machine learning, the Python AI ecosystem, data preparation, NLP, foundation models, and databases.

DAY

01

Day 1: Practical expectation of both IBM IJP roles

JD 1 is more AI solution architecture focused: LLMs, NLP, foundation models, Watson/IBM AI stack, and full AI lifecycle.

KEY COVERAGE Difference between JD 1 and JD 2; JD 1: Data Scientist - Artificial Intelligence; JD 2: Data Scientist - Advanced Analytics / GenAI / Databricks; Common skills required in both roles; Python

SOURCE [Day-1-IBM-IJP-Roles-Preparation.md](#) | PDF 21 pages

DAY

02

Day 2: Machine Learning Fundamentals

Machine Learning is a way to teach computers to learn patterns from data instead of writing fixed rules manually.

KEY COVERAGE What Machine Learning is; Main types of Machine Learning; Simple comparison table; Supervised Learning; Practical examples

SOURCE [Day-2-ML-Fundamentals-for-IBM-IJP.md](#) | PDF 28 pages

DAY

03

Day 3: Python AI Framework Ecosystem

Python is important in AI because most AI libraries, examples, models, and tools are built around Python.

KEY COVERAGE Big picture: Why Python is important in AI projects; Python AI ecosystem ASCII diagram; NumPy for numerical operations; What is NumPy?; Easy example

SOURCE [Day-3-Python-AI-Frameworks-Overview.md](#) | PDF 16 pages

DAY

04

Day 4: EDA Data Preprocessing Overview

EDA means understanding your data before building a model.

KEY COVERAGE What EDA is; Why EDA is important before ML; Understanding data shape, columns and data types; Data shape; Columns

SOURCE [Day-4-EDA-Data-Preprocessing-Overview.md](#) | PDF 26 pages

Core AI and data foundations

DAY

05

Day 5: NLP Fundamentals for IBM AI/GenAI Preparation

NLP means teaching computers to understand and work with human language.

KEY COVERAGE What NLP is; Structured vs unstructured data; Structured data; Unstructured data; Simple comparison

SOURCE [Day-5-NLP-Fundamentals-for-IBM-AI.md](#) | PDF 23 pages

DAY

06

Day 6 — Foundation Models and LLM Fundamentals

Foundation models are large AI models trained on huge datasets so they can be reused for many tasks.

KEY COVERAGE What foundation models are; What LLMs are; Traditional ML models vs foundation models; Easy example; Traditional ML

SOURCE [Day-6-LLM-Fundamentals-Overview.md](#) | PDF 18 pages

DAY

07

Day 7: SQL, NoSQL and Data Storage Concepts for AI Systems

Databases are where AI systems store raw data, processed data, documents, chunks, embeddings, user queries, and model responses.

KEY COVERAGE Why databases matter in AI projects; SQL basics; Tables, rows and columns; Easy SQL examples; Example 1: Select all data

SOURCE [Day-7-Databases-for-AI-Systems.md](#) | PDF 22 pages

RAG and agent workflows

Retrieval-augmented generation, embeddings, advanced retrieval, evaluation, LangChain, LangGraph, and multi-agent patterns.

DAY

08

Day 8: Retrieval-Augmented Generation, RAG

RAG means Retrieval-Augmented Generation.

KEY COVERAGE What RAG is; Why normal LLMs are not enough for enterprise knowledge; Problem 1: They do not know private company data; Problem 2: LLM knowledge may be outdated; Problem 3: LLMs can hallucinate

SOURCE [Day-8-RAG-for-Enterprise-Knowledge.md](#) | PDF 18 pages

DAY

09

Day 9: Embeddings and Vector Databases for RAG

Embeddings convert text into numbers so computers can understand meaning.

KEY COVERAGE What embeddings are; Why text is converted into numbers; Sentence embeddings; Similarity search; Cosine similarity

SOURCE [Day-9-Embeddings-and-Vector-Databases.md](#) | PDF 16 pages

DAY

10

Day 10: Advanced RAG Patterns

Basic RAG means: search documents, fetch chunks, send them to an LLM, and generate an answer.

KEY COVERAGE Why basic RAG may fail; Poor chunking problem; Good chunking principles; Missing context problem; Keyword search vs vector search

SOURCE [Day-10-Advanced-RAG-Patterns.md](#) | PDF 27 pages

DAY

11

Day 11: Evaluating and Improving a RAG System

RAG evaluation checks whether the system retrieved the right documents and generated the right answer.

KEY COVERAGE Why RAG evaluation is important; Retrieval quality; Answer quality; Groundedness; Faithfulness

SOURCE [Day-11-Evaluating-RAG-Systems.md](#) | PDF 22 pages

RAG and agent workflows

DAY

12

Day 12: LangChain Fundamentals

LangChain is a framework that helps developers build LLM-powered applications faster.

KEY COVERAGE What LangChain is; Why frameworks are used in GenAI apps; LLM wrapper; Easy example; Why useful?

SOURCE [Day-12-LangChain-Fundamentals.md](#) | PDF 15 pages

DAY

13

Day 13: LangGraph and Agentic AI

Agentic AI means an AI system can reason, choose steps, use tools, and continue a task until completion.

KEY COVERAGE What Agentic AI means; Difference between normal LLM app and agentic AI app; What LangGraph is; Nodes; Edges

SOURCE [Day-13-LangGraph-and-Agentic-AI.md](#) | PDF 16 pages

DAY

14

Day 14: CrewAI, Microsoft Semantic Kernel and Multi-Agent AI Patterns

A multi-agent system uses multiple AI agents, where each agent has a specific role.

KEY COVERAGE What multi-agent systems are; Why multiple agents are used; Planner agent; Researcher agent; Critic or reviewer agent

SOURCE [Day-14-Multi-Agent-AI-Overview.md](#) | PDF 17 pages

Data platform and production delivery

Databricks, Delta Lake, MLflow, MLOps, model APIs, cloud architecture, and an end-to-end enterprise GenAI solution.

DAY

15

Day 15: Databricks Fundamentals

Databricks is a cloud platform for data engineering, analytics, machine learning, and GenAI.

KEY COVERAGE What Databricks is; Why enterprises use Databricks; Enterprise benefits; Lakehouse concept; Data Lake

SOURCE [Day-15-Databricks-Fundamentals-Overview.md](#) | PDF 17 pages

DAY

16

Day 16: Delta Lake ETL ELT Guide

Delta Lake is a reliable storage layer used on top of a data lake.

KEY COVERAGE What Delta Lake is; Why normal data lakes have problems; ACID transactions; Schema enforcement; Schema evolution

SOURCE [Day-16-Delta-Lake-ETL-ELT-Guide.md](#) | PDF 22 pages

DAY

17

Day 17: MLflow for Machine Learning Lifecycle Management

MLflow is a tool to manage the full ML lifecycle: experiments, models, versions, deployment, and monitoring.

KEY COVERAGE What MLflow is; Why experiment tracking matters; Without MLflow; With MLflow; Parameters

SOURCE [Day-17-MLflow-for-ML-Lifecycle.md](#) | PDF 21 pages

DAY

18

Day 18: MLOps for Production AI Systems

MLOps means applying DevOps-style discipline to machine learning systems.

KEY COVERAGE What MLOps is; Why ML models need lifecycle management; CI/CD for ML; CI for ML; CD for ML

SOURCE [Day-18-MLOps-for-AI-Systems.md](#) | PDF 26 pages

Data platform and production delivery

DAY

19

Day 19 — Deploying ML and GenAI Models with REST APIs and Web Applications

A trained model becomes useful when other applications can send it data and receive predictions through an API.

KEY COVERAGE Why REST APIs are used for AI services; Request and response design; Example ML request; Example ML response; Recommended response fields

SOURCE [Day-19-Deploying-ML-Models-API.md](#) | PDF 31 pages

DAY

20

Day 20 — Cloud Platforms and Enterprise AI Architecture

Cloud platforms provide on-demand servers, storage, databases, networking and managed AI services without requiring companies to own physical infrastructure.

KEY COVERAGE Why cloud is important for AI systems; Main cloud benefits for AI; Elasticity; Managed services; Access to specialised hardware

SOURCE [Day-20-Cloud-AI-Architecture-Guide.md](#) | PDF 32 pages

DAY

21

Day 21 — End-to-End Enterprise GenAI Solution

The internal policy assistant retrieves relevant company documents before asking an LLM to generate an answer.

KEY COVERAGE Business problem; Proposed solution; Business benefits; Functional requirements; Non-functional requirements

SOURCE [Day-21-Enterprise-GenAI-Solution-Design.md](#) | PDF 46 pages

Find a topic fast

Use this cross-reference when you know the note topic but not the day that covers it.

Need to find...	Start with	Best note
Day 1: Practical expectation of both IBM IJP roles	Difference between JD 1 and JD 2	Day 01
Day 2: Machine Learning Fundamentals	What Machine Learning is	Day 02
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Day 4: EDA Data Preprocessing Overview	What EDA is	Day 04
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Day 11: Evaluating and Improving a RAG System	Why RAG evaluation is important	Day 11

Find a topic fast

Need to find...	Start with	Best note
Day 12: LangChain Fundamentals	What LangChain is	Day 12
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Day 15: Databricks Fundamentals	What Databricks is	Day 15
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Day 18: MLOps for Production AI Systems	What MLOps is	Day 18
Day 19 — Deploying ML and GenAI Models with REST APIs and Web Applications	Why REST APIs are used for AI services	Day 19
Day 20 — Cloud Platforms and Enterprise AI Architecture	Why cloud is important for AI systems	Day 20
Day 21 — End-to-End Enterprise GenAI Solution	Business problem	Day 21

COLLECTION SUMMARY

Source format	21 Markdown files in ljp/
Reading format	21 companion PDFs in ljp/pdfs/
Total coverage	27,682 source lines and 480 PDF pages
Mermaid inventory	0 Mermaid blocks in the generated collection
Math inventory	0 LaTeX math expressions in the generated collection